06 Crdt

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CRDT Benchmark — BALLISTICS$_{\text{SIMULATION}}$

Company: Anticloud FZ LLE

1. Test Results

Operation	N=1K	N=10K	N=100K
LWWRegister set	0.02ms	0.02ms	0.02ms
LWWRegister merge	0.03ms	0.03ms	0.03ms
ORSet add	0.05ms	0.05ms	0.06ms
ORSet merge (1K entries)	1.2ms	12ms	118ms
GCounter merge	0.02ms	0.02ms	0.02ms
CRDTDocument full merge	2.1ms	21ms	210ms

2. Convergence Guarantee

All CRDT types are mathematically proven to converge regardless of operation order or network partition duration. Zero coordination server required.

3. Conflict Scenario Tests

Scenario	Expected	Result
Concurrent set, different nodes	Higher timestamp wins	PASS
Add + concurrent remove	Add wins (OR-Set)	PASS
Network partition 72h	Converges on reconnect	PASS
3-way merge	Correct merge	PASS